1. Why we chose Git and how it is relevant to our project.

We are both 2 different students, not living together, and both of us work on different operating systems even. Aleksandar on Windows, and Efe on MacOS. We needed a fast way to synchronize any changes we make during the project, and easily resolve and code conflicts. That's where Git came, which allowed us to avoid sending zip files back and forth via email, and possibly naming the project a lot of different things.

2. What we learned through our research.

By reading guides online, namely Atlassian Git Tutorial (Atlassian, n.d.), and asking LLMs to simplify some concepts. For example, Aleksandar asked if it's best to directly merge our two different branches into the master branch, and received mixed opinions, and in the end made the decision that having a separate staging branch, where we could resolve any conflicts was the better choice. Only after everything went smoothly with the staging branch, did we merge it with the master branch.

The Atlassian Git tutorial helped us with understanding the concepts of a repository, branching, rebasing, commits, pushes to an online git repository (GitHub), and how to pull from a repository.

3. How we applied Git in our project.

We first created a repository in GitHub, and Aleksandar pushed his local Git folder as the first commit for the project. Then we, each, created our separate branches (Namely, "Aleksandar" and "Efe"). Then, after separating the workload between us, we started committing and pushing our individual parts. As we worked on different parts of the project, we didn't meet any merge conflicts until the end, when it was time to merge both of our branches together.

Furthermore, Aleksandar had to apply rebasing, as there were two nested repositories in the beginning, in order to fix a bigger future issue.

4. What we learned from the implementation, including challenges, decisions, and its impact on our project.

We faced a challenge in the very beginning, even. When we created the repository, Aleksandar pushed his local folder to serve as the basis for the project, but Efe thought he had to do it so he pushed his folder. Then Aleksandar pulled from the Repository, in order to get the newest base for the project, which then resulted in a nested git repository. This required manual work from Aleksandar in order to clean up the files, and remove that nested repo. This was done with git rebase. After that, the repository was fixed and Efe was notified to pull it, and then continue working on his respective part.

Then, we created our two separate branches, and worked without any issues until the end, when we had to merge. There seemed to be merged conflicts when Efe was finalising the UI with the logic of the game. He edited two files from the Jokers which were already done and pushed by Aleksandar, and this caused a conflict when we tried merging the two branches. However, that required a simple fix, and we were able to move from the staging branch to the master branch, and finish the project.

5. References

Atlassian. (*n.d*). *How to use Git? Tutorials, workflows & Commands | Atlassian*.

https://www.atlassian.com/git